EXPERIMENT-22

Database Connectivity Using PHP and MYSQL

1. Aim

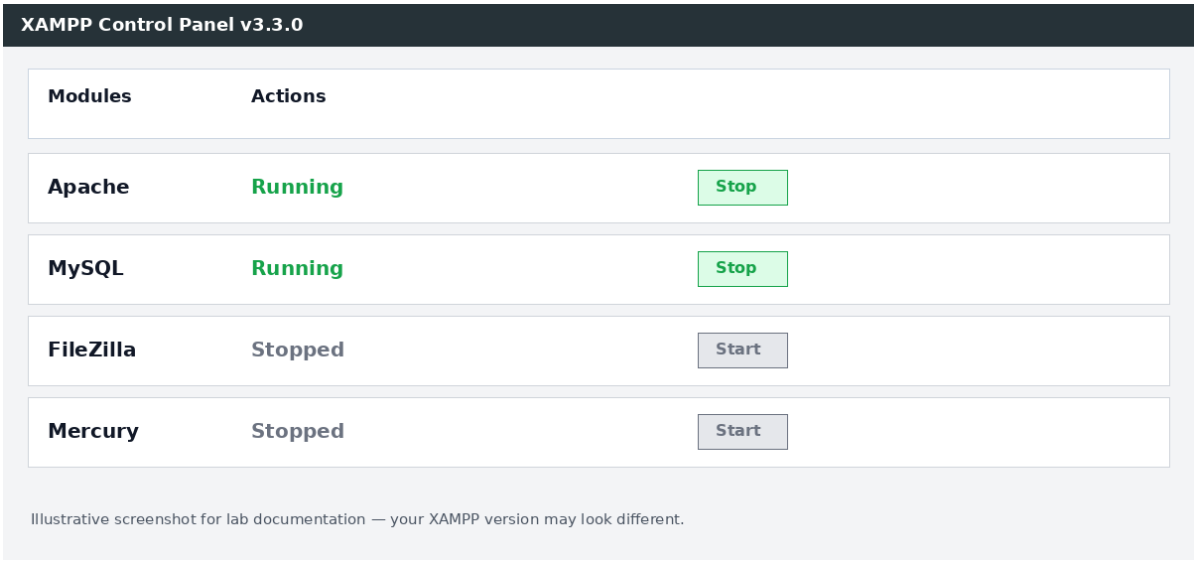
To establish a connection between PHP and a MySQL database, retrieve student records from a table, and display the records in a web browser.

2. Software Required

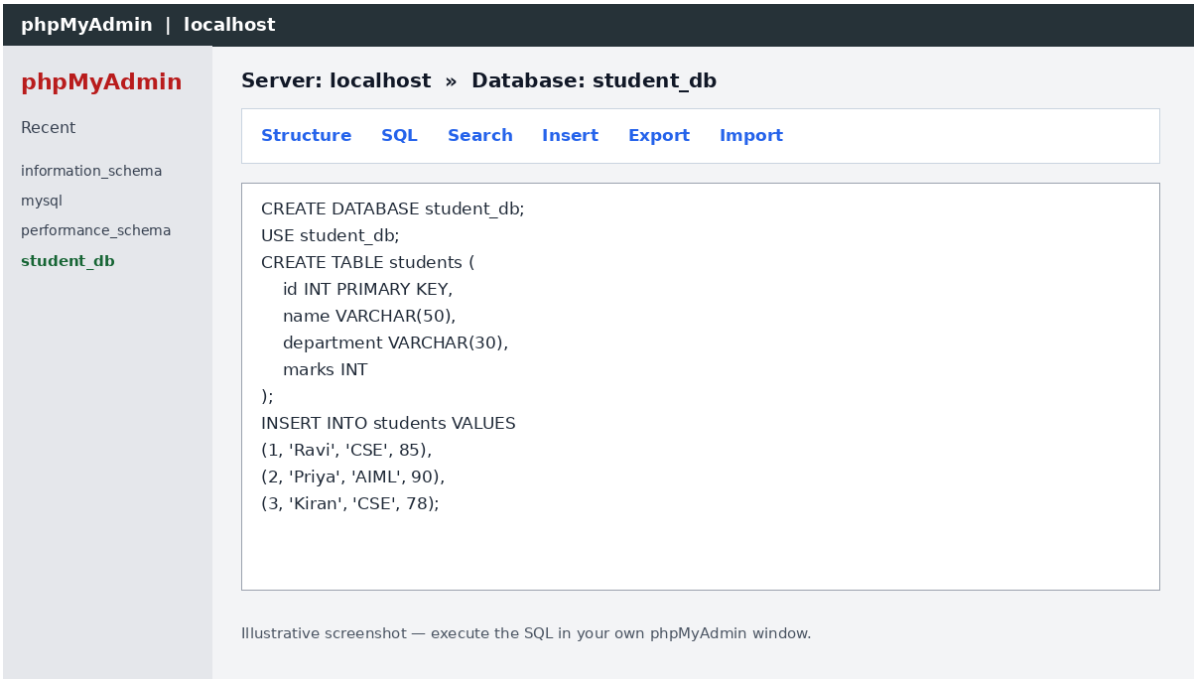
- XAMPP (Apache and MySQL/MariaDB)
- PHP with MySQLi extension (included with XAMPP)
- phpMyAdmin
- Visual Studio Code or any text editor
- Google Chrome or another web browser

3. Procedure

1. Install XAMPP and open the XAMPP Control Panel.
2. Start the Apache and MySQL modules.
3. Open http://localhost/phpmyadmin in a web browser.
4. Open the SQL tab and execute the database and table creation statements given below.
5. Create the folder C:\xampp\htdocs\student_connect.
6. Create a file named index.php in that folder and copy the PHP program below.
7. Open http://localhost/student_connect/index.php in the browser.
8. Verify the connection message and the student table displayed on the page.



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5. SQL Program

```
CREATE DATABASE student_db;

USE student_db;

CREATE TABLE students (
    id INT PRIMARY KEY,
    name VARCHAR(50),
    department VARCHAR(30),
    marks INT
);

INSERT INTO students VALUES
(1, 'Ravi', 'CSE', 85),
(2, 'Priya', 'AIML', 90),
(3, 'Kiran', 'CSE', 78);

SELECT * FROM students;
```

6. PHP Program (index.php)

```
<?php
$servername = "localhost";
$username = "root";
$password = "";
$dbname = "student_db";

$conn = mysqli_connect($servername, $username, $password, $dbname);

if (!$conn) {
    die("Connection failed: " . mysqli_connect_error());
}
```

```
echo "<h2>Database Connected Successfully</h2>";

$sql = "SELECT * FROM students";
$result = mysqli_query($conn, $sql);

echo "<h3>Student Details</h3>";

if (mysqli_num_rows($result) > 0) {
    echo "<table border='1' cellpadding='10'>";
    echo "<tr><th>ID</th><th>Name</th><th>Department</th><th>Marks</th></tr>";

    while ($row = mysqli_fetch_assoc($result)) {
        echo "<tr>";
        echo "<td>" . $row['id'] . "</td>";
        echo "<td>" . $row['name'] . "</td>";
        echo "<td>" . $row['department'] . "</td>";
        echo "<td>" . $row['marks'] . "</td>";
        echo "</tr>";
    }

    echo "</table>";
} else {
    echo "No records found";
}

mysqli_close($conn);
?>
```



Figure 3: Illustrative PHP editor view. The complete executable program is printed above.

7. Execution Steps

9. Open XAMPP Control Panel and start Apache and MySQL. The module status should show Running.

10. Open phpMyAdmin at http://localhost/phpmyadmin and execute the SQL statements.
11. In File Explorer, navigate to C:\xampp\htdocs and create the folder student_connect.
12. Save the PHP program as C:\xampp\htdocs\student_connect\index.php. Ensure the filename does not end in .txt.
13. Open Chrome and enter http://localhost/student_connect/index.php.
14. If the program runs correctly, the browser displays the connection message and student details.

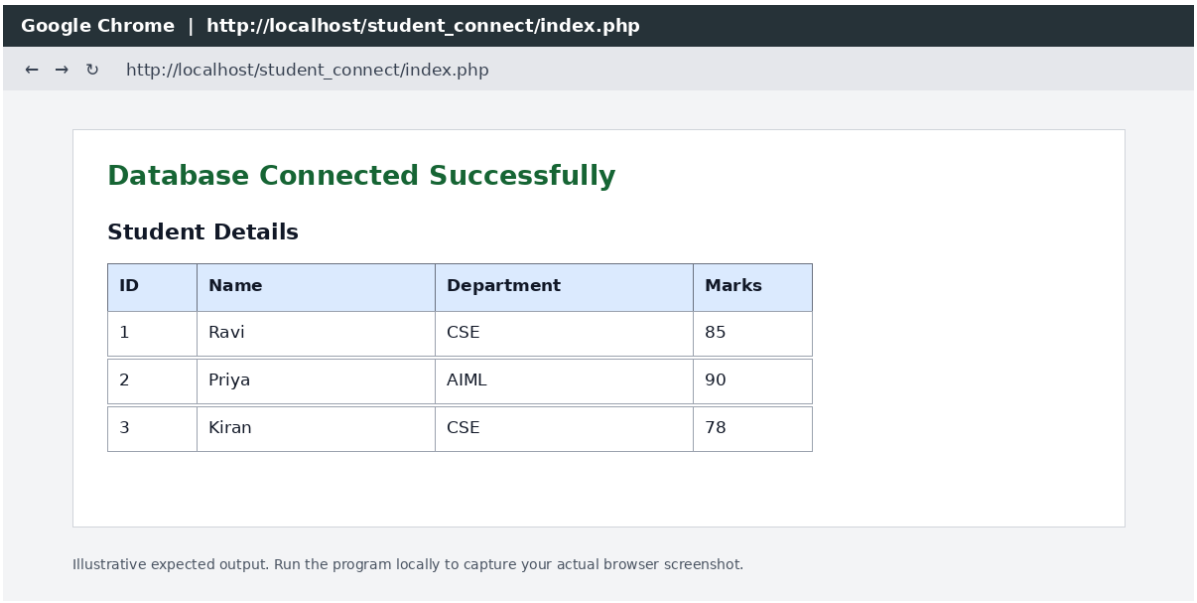


Figure 4: Illustrative expected browser output with the connection message and student records.

Note: The figures in this document are illustrative screenshots created to show the expected interface and output. They are not captured from your computer. After running the program, you can take actual screenshots using Win + Shift + S and replace these figures if your lab requires proof of execution.

8. Sample Output

Database Connected Successfully

Student Details

ID	Name	Department	Marks
1	Ravi	CSE	85
2	Priya	AIML	90
3	Kiran	CSE	78

9. Explanation of Important Functions

Function	Purpose
mysqli_connect()	Connects PHP to the MySQL database server and selected database.
mysqli_query()	Executes the SQL SELECT query.
mysqli_num_rows()	Checks how many records were returned.

mysqli_fetch_assoc()	Fetches one row as an associative array using column names.
mysqli_close()	Closes the database connection.

10. Troubleshooting

Problem	**Solution**
Connection failed / Access denied	Confirm MySQL is running and use the correct username and password. The default XAMPP root account often has a blank password on a local installation.
Unknown database student_db	Run the CREATE DATABASE and USE statements in phpMyAdmin.
Table students doesn't exist	Run the CREATE TABLE and INSERT statements.
404 Not Found	Check the folder and filename under C:\xampp\htdocs\student_connect.
PHP code appears as plain text	Start Apache and use the localhost URL instead of opening the file directly.
Port conflict	If Apache or MySQL will not start, check whether another program is using the configured port.

Result

Thus, database connectivity between PHP and MySQL was demonstrated successfully. The program connected to the student_db database, retrieved student records from the students table, and displayed them in the web browser.